Windows Server Assignment

Module 1:

Windows Server 2016Installation Requirements:

- Processor: 1.4 GHz 64-bit
- RAM: Minimum 512 MB (2 GB recommended)
- Disk Space: Minimum 32 GB
- Network Adapter: Ethernet

Windows Server 2016 Editions:

- Datacenter Edition (advanced features like virtualization)
- Standard Edition (basic features)
- Essentials Edition (small businesses)

Add/Remove Server Roles:

- Server Manager → Manage → Add Roles and Features

Workgroup:
A peer-to-peer network without centralized control.

Domain:
A centralized network managed by a Domain Controller.

PowerShell:
A command-line shell and scripting language for automation.

Upgrade vs Migration:

- Upgrade: Install new version over old OS
- Migration: Move roles/data to new server

License & Activation Model:

- Based on cores and CALs

Precautions of Upgrade:

- Backup data

- Check compatibility
- Verify hardware requirements

Migration Limitations:

- Time-consuming
- Application compatibility issues

Server Core Advantages:

- Less resource usage
- Reduced attack surface
- Better performance

Nano Server:

- Lightweight installation option
- Designed for cloud and containers

Purpose of Nano Server:

- Run cloud-native apps
- Minimal footprint

Comparison:

- GUI: User-friendly, heavy
- Core: No GUI, efficient
- Nano: Ultra-lightweight

Practical Tasks

- Assign dual IP: Configure via Network Adapter settings
- Upgrade Server 2012 → 2016
- Change computer name: System Properties
- Install & manage Nano Server using PowerShell

Storage Solutions

GPT vs MBR:

- GPT: Modern, supports large disks
- MBR: Older, limited to 2 TB

VHD vs VHDX:

- VHDX supports larger size and better performance

SMB & NFS:

- SMB: Windows file sharing
- NFS: Linux/Unix file sharing

Permissions:

- Sharing Permission: Network level
- NTFS Permission: File system level

Storage Concepts:

- Storage Pool: Group of disks
- Basic Disk: Standard partition
- Dynamic Disk: Advanced volumes

Volumes:

- Simple Volume: Single disk
- Spanned Volume: Multiple disks

RAID Levels:

- RAID 0: Striping (no redundancy)
- RAID 1: Mirroring
- RAID 5: Parity
- RAID 6: Dual parity
- RAID 10: Mirror + striping

Storage Types:

- DAS: Direct Attached Storage
- NAS: Network storage

- SAN: High-speed network storage

Module 2: Networking

DNS

- Resolves domain names to IP addresses
- Query types:
    - Recursive
    - Iterative

Zones:

- Forward Lookup Zone
- Reverse Lookup Zone

Records:

- A, AAAA, MX, CNAME

DHCP

Purpose:

Automatically assign IP addresses

DORA Process:

- Discover
- Offer
- Request
- Acknowledge

Features:

- Scope
- Lease
- Reservation
- Failover

IPAM

IP Address Management:

- Tracks IP usage
- Manages DHCP & DNS

VPN & Remote Access

VPN:

Secure connection over internet

Types:

- PPTP
- L2TP
- SSTP

Network Policy Server (NPS)

- Acts as RADIUS server
- Provides authentication, authorization, accounting

IP Addressing

- IPv4 and IPv6
- Public vs Private IP
- NAT
- Gateway
- Loopback

DFS

Distributed File System:

- Namespace
- Replication

Advanced Networking

- SDN: Software Defined Networking
- SCVMM: Virtual machine management

Module 3: Active Directory

AD DS Concepts

- Domain Controller
- Forest, Tree, Domain
- OU (Organizational Unit)
- Schema

Global Catalog:

Stores searchable directory data

FSMO Roles

- Schema Master
- Domain Naming Master
- RID Master
- PDC Emulator
- Infrastructure Master

AD Tasks

- Install AD DS
- Create users/groups
- Join domain
- Configure RODC & ADC

Advanced AD Features

- Trust Relationships

- Sites & Subnets
- Backup & Restore
- Replication tools

Features:

- GPO linking
- Filtering
- Scripts
- Software deployment

Certification Services

- Provides digital certificates
- Types:
    - Standalone CA
    - Enterprise CA